

Adaptive Ultralocalized Time-Series for Improved Model-Free Predictive Current Control on PMSM Drives

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Abstract—Since a data-driven model is adopted to describe the operating state of the plant in the model-free predictive control, it has been widely used in the motor driving realm to eliminate the influences caused by parameter mismatches and enhance the robustness of the system. However, due to the fixed model structure and heavy calculating process, it is difficult to obtain an improved control performance using time-series models in continuous-control-set (CCS) predictive algorithms. To solve these problems, a model-free predictive current control (MF-PCC) using adaptive ultralocalized time-series is proposed in this article, and applied to a permanent magnet synchronous motor driving system as the current controller. The model structure is improved as a variable, and its orders are online adjusted according to the designed adaptive law and the current operating state of the system. The complex discrete-time transfer functions in the model are ultralocalized to simplify the realization in the CCS-type controller. All required coefficients in the model are estimated by the recursive least squares algorithm, and the optimal gain is also found by the particle swarm optimization algorithm. The effectiveness of the proposed method is demonstrated by the experimental results, as well as the advantages of the proposed method, including better model accuracy and current quality with suitable robustness compared with the conventional time-series based MF-PCC.

Index Terms—Adaptive ultralocalized time-series (AULTS), continuous-control-set (CCS) type, data-driven model, model-free predictive current control (MF-PCC).

I. INTRODUCTION

PERMANENT magnet synchronous motor (PMSM) is widely used in industry and equipment due to its advantages, such as high power density, small volume, and light weight [1]. With the development of PMSM applications, the control requirements become more strict to adapt to demanding operation conditions and hostile environments, such as time-varying physical parameters, magnetic field coupling, and iron core saturation, which pose threats to control accuracy and reliability. Since the traditional controllers cannot satisfy these requirements fully, with the evolution of the digital processor, model predictive control (MPC) has raised interest in the motor driving realm due to its excellent dynamics, rolling optimization characteristics, and multiobjective optimization [2]. MPC can be mainly divided into three types in this realm, including predictive current control (PCC), predictive speed control, and predictive torque control according to the primary objective, and also can fall into finite-control-set (FCS) type and continuous-control-set (CCS) type depending on the realization of MPC [3], [4]. A discrete-time a priori model of the plant is necessary to predict the variables in the conventional MPC, and the model accuracy directly affects the prediction error and the control performances [5]. However, because of the influences caused by the time-varying physical parameters in the motor and inevitable parameter mismatches in the a priori model, weak robustness is a serious problem limiting the high-end applications of MPC [6].

Researchers address this problem from multiple perspectives, such as identification-based predictive control, observer-based predictive control, and model-free predictive control (MFPC) [7], [8]. The first one is used to online correct the physical parameters of the motor in the a priori model and thus, reduce the influences caused by the mismatches, such as the stator inductance identification by an improved model reference adaptive system based on fuzzy logic in [9] or by nonintrusive parasitic transient tracking errors independent to other disturbances of the high-speed motor in [10]. Considering multiple physical parameters at the same time, an interesting idea of differential

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parameter maps for online self-commissioning was presented in [11] in an optimized MPC with long-term memory recursive least squares (RLS) current estimator with good dynamics. For the observer-based predictive control, all internal and external disturbances, including the influences caused by mismatches, are calculated and compensated to the variables by the observers, such as the disturbance observer in [12], extend state observer in [13], and sliding mode observer in [14]. Some advanced observers are presented to obtain a precise disturbance, such as the predictive error model-based enhanced observer in [15] achieving strong robustness, great stability, and convergence.

Although the robustness is effectively enhanced in the above-mentioned two methods, since the a priori model is involved, they belong to MPC and the influences caused by parameter mismatches still exist. Based on the consideration of model-free control, to essentially eliminate these influences, the a priori model is replaced by a data-driven model in MFPC, which is only related to the sampled data [16], [17]. Many kinds of data-driven models can describe the motor driving system, e.g., ultralocal, time-series, and look-up table (LUT) models. The ultralocal is the most commonly used model in MFPC due to its simple structure and easy realization, all terms from the plant are summarized as a single variable, and this variable is estimated by some observers [18], [19]. This model is achieved in CCS-type which is straightforward due to the proportional-integral structure of the model, and the model accuracy can be improved by the reductions of the effects of harmonic coupling and side-frequency effects in [20] or considerations of high model orders and affine terms to show the nonlinear and high-dimension motions in [21] and [22]. Time-series models have multiple structures, for instance, autoregressive with exogenous input in [23] and autoregressive moving average in [8], which can be selected according to the provided data type. This modeling approach represents the plant as discrete-time transfer functions, and the coefficients in the model are online selected by an estimation algorithm. To alleviate the issue of the hard realization in CCS-type, the reversed time-series model and the control law was skillfully designed by the time-shift with the Lagrange algorithm without any extra calculation burden in [24]. For the LUT model, the typical data-driven model in MFPC is abandoned, and replaced by LUTs storing variable gradients caused by each candidate vector, where all information in LUTs should be updated in time to ensure stability [25]. The 1-D LUT is expanded into 2-D LUT in MFPC to consider the effects of the rotor position and eliminate current spikes under the conditions of field-weakening [26]. The stagnation effect caused by low updating frequency is the primary issue for the accuracy and updating of LUT model. To address this issue, more vector selection processes are considered to improve the current quality, such as double-vector in [27] and [28] and triple-vector in [29]. By a second-order generalized integrator filter, updating the intermediate terms to reconstruct all current variations, the stagnant variation update was also removed effectively in [30].

Moreover, dynamic linearization and neural networks are also reliable models to represent the system, in which the former approximates the plant as either compact-format, particle-format,

or full-format based on the pseudopartial derivative principle, and is commonly used in model-free predictive adaptive control [31]. The latter expresses the plant as a neural network model with an unfixed structure and high flexibility [32].

It is difficult to apply multiple model ideas with better flexibility in [33] in MFPC to upgrade model accuracy because of limited processor resources. To address this issue, an improved model-free PCC (MF-PCC) strategy that leverages an automatically changing structure of data-driven models is proposed in this article, for improving the accuracy of the time-series model in CCS-type and optimizing gains in PMSM driving systems. The strategy uses an adaptive ultralocalized time-series (AULTS) model to describe the plant, which estimates coefficients online with the RLS algorithm and updates orders using an adaptive law to maintain high model accuracy. The model of a group of complex discrete-time transfer functions is ultralocalized, allowing for easy realization in CCS-type, and the optimal gain is found using the particle swarm optimization (PSO) algorithm. Simulation and experimental results demonstrate that the AULTS model is effective, and the proposed MF-PCC strategy improves model accuracy and current quality with good robustness compared to conventional MF-PCC and CCS-PCC strategies. To summarize, this article has the following main contributions.

- 1) In order to design a time-series-based MF-PCC strategy in the CCS-type, this article proposes an ultralocalization method that skillfully adjusts the model structure as an AULTS model and easily implements the strategy in the CCS-type based on a suitable control law.
- 2) An adaptive model order selection method is designed for the time-series model to online improve the model flexibility and accuracy to adapt to the current operating state of the system.
- 3) The stability and robustness become weak in the ultralocal model and ultralocalized model if the gain in control law is not properly designed. A PSO algorithm is adopted in the proposed method to automatically select the optimal value of this gain, to prevent this drawback.

The rest of this article is organized as follows. Section II introduces the a priori model and two typical data-driven models of PMSM. Section III presents the improved MF-PCC strategy using AULTS and its estimation and optimization algorithms for the coefficients and gain respectively. Experimental results are demonstrated in Section IV to validate the proposed method and its advantages. Finally, Section V concludes this article.

II. PMSM MODEL AND TYPICAL PCC STRATEGIES

A. PMSM Model

On the dq coordinate frame, PMSM is expressed as the following state functions of stator voltage u_s , stator flux linkage ψ_s , and electromagnetic torque T_e

$$\begin{cases} u_{sd} = R_s i_{sd} + L_{sd} \frac{di_{sd}}{dt} - L_{sq} \omega_r i_{sq} \\ u_{sq} = R_s i_{sq} + L_{sq} \frac{di_{sq}}{dt} + L_{sd} \omega_r i_{sd} + \omega_r \psi_m \end{cases} \quad (1)$$

$$\begin{cases} \dot{\psi}_{sd} = L_{sd} i_{sd} + \psi_m \\ \dot{\psi}_{sq} = L_{sq} i_{sq} \end{cases} \quad (2)$$

$$T_e = p[\psi_m i_{sq} + (L_{sd} - L_{sq}) i_{sd} i_{sq}] \quad (3)$$

where the subscripts d and q are the components on d -axis and q -axis of a synchronous reference frame respectively, R_s is the stator resistance, L_s is the stator inductance, ψ_m is the magnet flux linkage, p is the number of pole pairs, and ω_r is the electrical angular velocity.

B. Typical CCS-PCC Strategies

Based on the sampling period T_s , the model in (1) is discretized by the forward Euler method to predict the stator current to the $(k+1)$ th sampling period, and the cost function is designed as follows to make the current components operate close to their references:

$$J = (i_{sd}(k+1))^2 + \lambda (i_{sq}^* - i_{sq}(k+1))^2 \quad (4)$$

where i_{sq}^* is the reference for i_{sq} , and λ is the weighting factor. The two-step Euler method and the Runge-Kutta method can be considered as replacements for the forward Euler method to achieve higher precision or higher-order discretization in predictive control.

In the CCS-type, to find the stator voltage with the minimal cost function value, the partial derivative with respect to the stator voltage satisfies the following condition:

$$\frac{dJ}{du_{sd}} = 0, \frac{dJ}{du_{sq}} = 0 \quad (5)$$

and the stator voltage is expressed as follows to generate pulses to control motor driver:

$$\begin{cases} u_{sd}(k+1) = \left(R_s - \frac{L_{sd}}{T_s}\right) i_{sd}(k) - L_{sq} i_{sq}(k) \omega_r \\ u_{sq}(k+1) = \left(R_s - \frac{L_{sq}}{T_s}\right) i_{sq}(k) + L_{sd} i_{sd}(k) \omega_r \\ \quad + \psi_m \omega_r + \frac{L_{sq}}{T_s} i_{sq}^* \end{cases} \quad (6)$$

For the MF-PCC, the a priori model of the plant is replaced by the data-driven model, such as the following typical one-order ultralocal and time-series models. All physical parameters are eliminated in the model.

$$\frac{di_{sx}}{dt} = \gamma u_{sx} + F_x \quad (7)$$

$$\begin{aligned} & \sum_{i=0}^n (\alpha_i z^{-i}) i_{sx}(k) - \sum_{j=0}^m (\beta_j z^{-j}) u_{sx}(k) = 0 \\ & \Rightarrow u_{sx}(k) \\ & = \frac{\alpha_n z^{-n} + \alpha_{n-1} z^{-(n-1)} + \dots + \alpha_1 z^{-1} + \alpha_0}{\beta_m z^{-m} + \beta_{m-1} z^{-(m-1)} + \dots + \beta_1 z^{-1} + \beta_0} i_{sx}(k) \end{aligned} \quad (8)$$

where the subscript x means d or q of the component, γ is the input gain, and F_x is the unknown part for the ultralocal model, α_i and β_j are the model coefficients (or discrete-time transfer function coefficients), and n and m are the model orders. α_0 is selected as 1 to ensure causality.

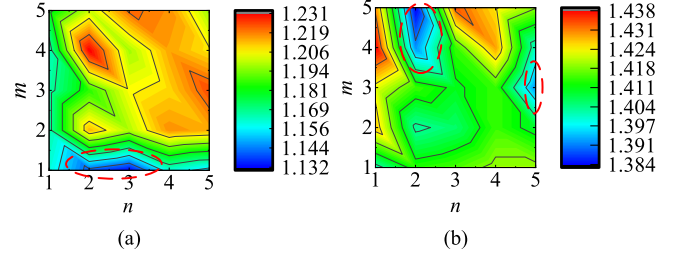


Fig. 1. Accumulated current error for MF-PCC strategy using time-series with different model orders during transient-state. (a) Case I. (b) Case II.

III. PROPOSED MF-PCC STRATEGY

A. Problem Description

The proposed method focuses on improving the control performance from three aspects: 1) the realization of the MF-PCC using a typical time-series model is hard in the CCS-type because partial derivatives and matrix inversions are difficult to realize in the digital processor; 2) the fixed orders for the model limit the model accuracy; and 3) a suitable input gain, which is illustrated in [16], is necessary to realize required performances. To highlight the second aspect, under the conditions of different transient-states in simulations, involving various fixed model orders in the MF-PCC using time-series, the accumulated current errors are shown in Fig. 1, in which data from two transient responses with changing reference and load torque are considered and labeled as Cases I and II, respectively. During the normal operating states, as the formulations in [23], the model accuracy is improved with the increasing orders. However, as shown in the figures, the optimal orders circled in red are not constant during the transient-states, even more than one group of suitable values exists. Therefore, the data-driven model with fixed orders cannot accurately reflect the current operating state of the plant, but rather increases the computational burden caused by excessive orders.

B. AULTS-Modeling and Updating Processes

The time-series model of PMSM with adaptive orders is

$$\sum_{i=0}^{n(k)} (\alpha_i z^{-i}) i_{sx}(k) - \sum_{j=0}^{m(k)} (\beta_j z^{-j}) u_{sx}(k) = 0 \quad (9)$$

where $n(k) \in \mathbb{Z}^+$ and $m(k) \in \mathbb{Z}^+$ are the discrete-time variables of model orders adjusted within each sampling period, and α_0 is also selected as 1.

All coefficients and sampled data are grouped as the following vectors θ_x and ψ_x respectively with the dimension of $[(n(k) + 1) + (m(k) + 2)]$ to make a preparation for the order selection process, and these coefficients are online estimated by the RLS algorithm.

$$\begin{cases} \psi_x(k) = [i_{sx}(k), \dots, i_{sx}(k - (n(k) + 1)), \\ u_{sx}(k), \dots, u_{sx}(k - (m(k) + 1))]^T \\ \theta_x(k) = [\alpha_0, \dots, \alpha_{n(k)+1}, \beta_0, \dots, \beta_{m(k)+1}]^T \end{cases} \quad (10)$$

The predictive current is obtained by the following time-shift, in which the predictive voltages are calculated by the Lagrange algorithm:

$$\hat{i}_{sx}(k+1) = \psi_{x,norm}^T(k+1)\hat{\theta}_{x,norm}(k) \quad (11)$$

where the superscript wedge means the estimated value, and

$$\begin{cases} \psi_{x,norm}(k) = [i_{sx}(k), i_{sx}(k-1), \dots, i_{sx}(k-n(k)), \\ u_{sx}(k), \dots, u_{sx}(k-m(k))]^T \\ \hat{\theta}_{x,norm}(k) = [\alpha_0, \hat{\alpha}_1, \dots, \hat{\alpha}_{n(k)}, \hat{\beta}_0, \dots, \hat{\beta}_{m(k)}]^T. \end{cases} \quad (12)$$

When α_0 is set as 1, rearranging (9) to the AULTS structure, and iterating it to time $k+1$ and evaluating the future current $i_{sx}(k+1)$ in the Euler's approximation of the derivative

$$\begin{aligned} \hat{i}_{sx}(k+1) &= \frac{1}{T_s} \left[(\hat{\alpha}_1 - 1) i_{sx}(k) + \sum_{i=2}^{n(k)} (\hat{\alpha}_i z^{-i}) i_{sx}(k+1) \right. \\ &\quad \left. - \sum_{j=1}^{m(k)} (\hat{\beta}_j z^{-j}) u_{sx}(k+1) \right] + \frac{\hat{\beta}_0}{T_s} u_{sx}(k+1) \end{aligned} \quad (13)$$

in which the first term is seen as the unknown part compared with (8)

$$\begin{aligned} \hat{F}_x(k+1) &= \frac{1}{T_s} \left[(\hat{\alpha}_1 - 1) i_{sx}(k) + \sum_{i=2}^{n(k)} (\hat{\alpha}_i z^{-i}) i_{sx}(k+1) \right. \\ &\quad \left. - \sum_{j=1}^{m(k)} (\hat{\beta}_j z^{-j}) u_{sx}(k+1) \right] \end{aligned} \quad (14)$$

and the second term is seen as the estimated input gain γ

$$\hat{\gamma}(k+1) = \hat{\beta}_0/T_s. \quad (15)$$

Combining with the proportional structure, the voltage control signal is generated by the following function to generate the pulses from a space vector modulation (SVM) and achieve in the CCS-type.

$$v_{sx}^* = \frac{1}{\hat{\gamma}} \left(-\hat{F}_x + \dot{i}_{sx}^*(k+1) \right) + \frac{k_p}{\hat{\gamma}} (i_{sx}^*(k+1) - i_{sx}(k+1)) \quad (16)$$

where k_p is the proportional gain.

C. Adaptive Order Selection

Since the current error between the predictive value and the sampled value directly reflects the model accuracy, an adaptive logic is designed to select a group of suitable orders.

To fully use the existing RLS algorithm, the dimensions of regressive vectors are adjusted in turn as follows with different orders of $[n(k)+1, m(k)]$, $[n(k), m(k)+1]$, $[n(k)-1, m(k)]$,

TABLE I
REQUIRED PARAMETERS IN MODELING PROCESS OF MF-PCCs

	A priori model	Ultra-local model in [13]	Time-series model in [24]	AULTS model
Required parameters	$R_s, L_{sd}, L_{sq}, \psi_m$	$\gamma (= 1/L_s), \omega^a$	m, n, λ^b	λ

^a ω means the bandwidth of the controller in [13].

^b λ means the forgetting factor in RLS algorithm.

and $[n(k), m(k)-1]$, respectively.

$$\begin{cases} \psi_{x,y}(k) = [i_{sx}(k), \dots, i_{sx}(k-n(k)+C_1), \\ u_{sx}(k), \dots, u_{sx}(k-m(k)+C_2)]^T \\ \hat{\theta}_{x,y}(k) = [\hat{\alpha}_0, \dots, \hat{\alpha}_{(n(k)+C_1)}, \hat{\beta}_0, \dots, \hat{\beta}_{(m(k)+C_2)}]^T \end{cases} \quad (17)$$

where the subscript y means a label of $[norm, ii, iv, di, dv]$, in which these labels reflect the changing ways of the vectors as shown in Fig. 2. C_1 and C_2 are the constants of 1, -1, or 0 determining the dimensions of the vectors.

To select a group of suitable orders of $n(k+1)$ and $m(k+1)$, all regressive vector contents with possible orders and the designed adaptive logic are shown in Fig. 2, in which e_{norm} , e_{ii} , e_{iv} , e_{di} , and e_{dv} are

$$e_y = \sum_{x=d,q} |i_{sx}^*(k+1) - \psi_{x,y}^T(k+1)\hat{\theta}_{x,y}(k)|. \quad (18)$$

The required parameters in the modeling and updating processes of some MF-PCC strategies are listed in Table I.

D. Gain Selection Using PSO

To maintain good universality and enhanced robustness for the model-free idea, and provide ample potential for future applications in the field of power electronics or other realms, when (16) is re-devised as a high-dimensional function to suit more complex applications, a PSO is designed to find the optimal value of the proportional gain k_p , and make the particles reflect the possible value of gain. The static errors of current are used to appraise every candidate particle, and the objective function is designed as follows combined with the consideration in [34]:

$$g_{opt} = \min(M + N + |N|) \quad (19)$$

where $M = \text{rms}(\Delta i_{sd}) + \delta \text{rms}(\Delta i_{sq})$ and $N = \text{rms}(\Delta i_{sq}) + \delta \text{rms}(\Delta i_{sd})$, in which δ is the factor ensured the converging error of the q -axis current. The first term of M is the error adjusting term to improve current quality, and the second term of $N + |N|$ is applied to reduce the q -axis current error.

Since the updating rate of k_p does not need to be as fast in the proposed method and PSO is a slow convergence process requiring a large setting time, the optimal gain is set to (16) after each continuous 200 sampling period to realize a suitable proportional gain.

Summarizing the abovementioned processes, the flowchart of the proposed method is shown in Fig. 3. Fig. 4 shows the structure of the current control part of the system, which involves the proposed method. The sampled data is used to estimate the model coefficients and adjust model orders in turn, to update the AULTS model. Combined with the optimized k_p from PSO and

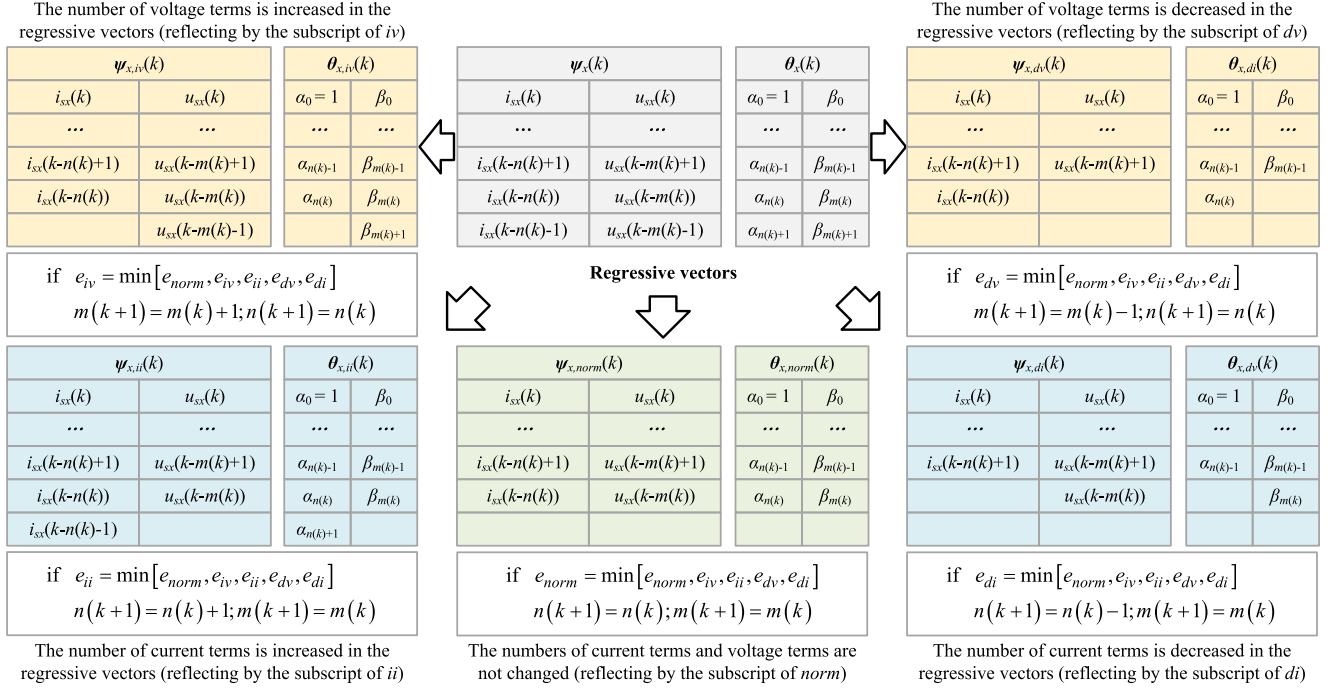


Fig. 2. Regressive vector contents and adaptive logic with all possible orders.

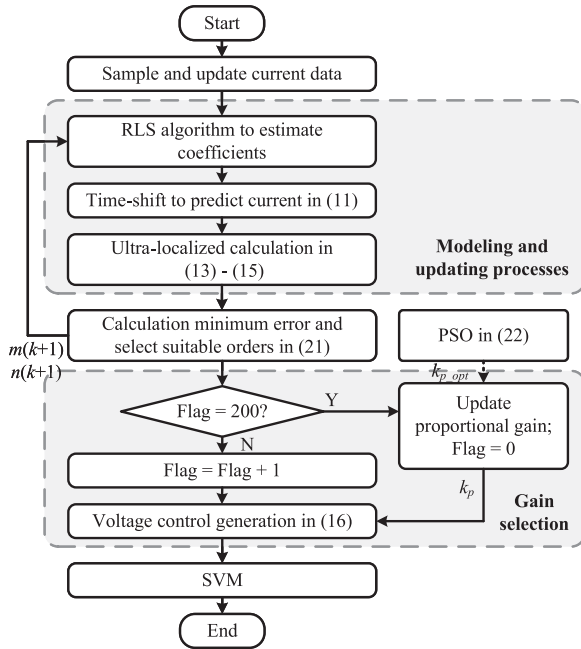


Fig. 3. Flowchart of the proposed method.

predictive currents from the model, the control law generates control variables sent to SVM to control the driver and realize objectives.

E. Stability Discussions

According to [35] and [36], the plant can be linearized as an inertial term, and the current control part of the system is

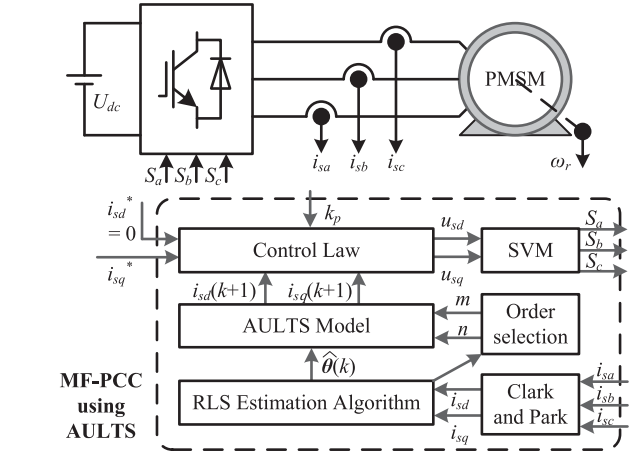


Fig. 4. Structure of the proposed method.

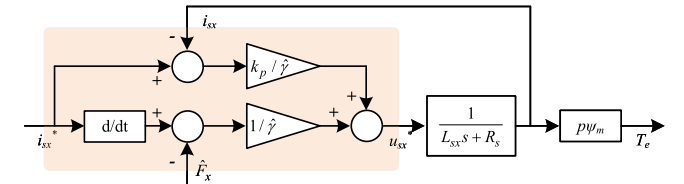


Fig. 5. Modeled current control part of the system.

modeled in Fig. 5. The model coefficients are sampled four times, and the variables of γ and F_x in (16) are fixed. The pole/zero map of the system is shown in Fig. 6. As shown in the figure, after simplification, the system only has one pole and one zero point. Under different operating conditions, zeros have

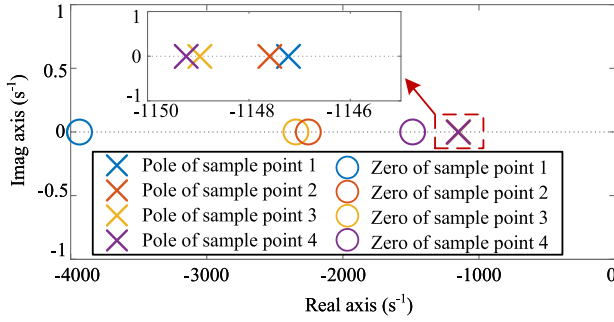


Fig. 6. Pole and zero map with different sample points.

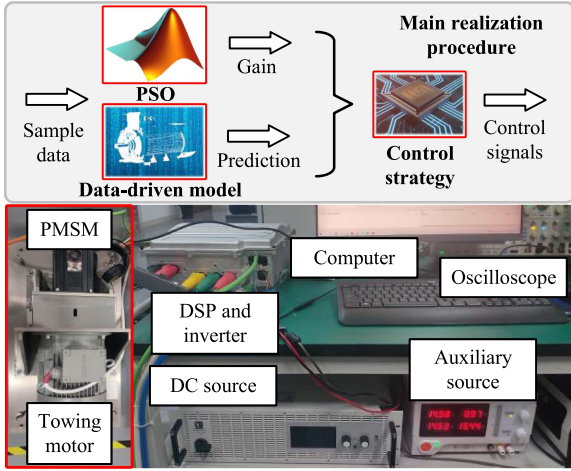


Fig. 7. Experimental platform.

a high sensitivity compared with the poles relating to four sample points. No matter how the zeros and poles of the system move, the poles are always on the negative real axis, and the system remains stable.

Moreover, according to the squeeze idea in [8], if a more accurate model is adopted in the proposed method, the prediction error will be reduced, and due to the stable rolling optimization process in the conventional method, stability can also be confirmed indirectly, under identical conditions in other parts.

IV. VALIDATIONS

The experimental platform is shown in Fig. 7, consisting mainly of a DSP F28379D, a 4.8 kW PMSM towing motor, an inverter (a voltage-source-type three-phase bridge-type inverter), a dc source, an auxiliary source, and an upper computer. The towing motor is driven by an ABB drive to generate a load torque during the operating process. Besides observing the electrical variables in real time with an oscilloscope, they are also sent to the computer via the serial communication interface on the DSP. The program of the control strategy is burned into DSP to verify the realized performance of the system, and according to the sampled currents, the PSO algorithm is calculated on the computer to obtain the gain and sent to the DSP to update the control law at the end of each 200 sampling periods. The main parameters of the platform are listed in Table II.

TABLE II
MAIN PARAMETERS OF THE SYSTEM

	Quantity and symbols	Values
Motor parameters	Stator resistance R_s	0.147 Ω
	Stator inductance on q -axis L_{sq}	14.5 mH
	Stator inductance on d -axis L_{sd}	10 mH
	Number of pole pairs p	4
	Magnet flux linkage ψ_m	0.31 Wb
	Rotor inertia J	0.005 kg.m ²
	DC voltage V_{dc}	540 V
Parameters for the proposed method	Sampling period T_s	62.5 μ s
	Forgetting factor λ	1
	Upper bound of orders	3
	Max number of iterations	200
	Max inertia weighting factor	0.9
	Min inertia weighting factor	0.2

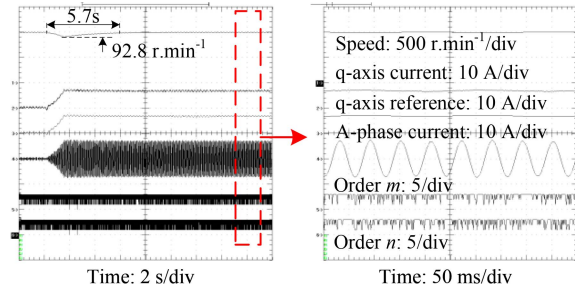
A. Basic Performances

The experimental waveforms of the orders, speed, A-phase current, q -axis current, and its reference during the disturbance rejection are shown in Fig. 8(a) for the proposed method, and the corresponding average value of orders is shown in Fig. 8(b), in which the average value is calculated based on the sample point group including the continuous 100 sample points to observe the changing orders easily. As shown in the figure, when the load torque is increased from zero to 11.5 N.m, the amplitude of the phase current is also increased to generate a larger electromagnetic torque, and the system resists the disturbance within about 5.7 s with a dropped depth of 9.28%. The average values of orders m and n are decreased at first, and the previous order is recovered last according to the tendency in Fig. 8(b), and this means the model does not require large orders over a period of time.

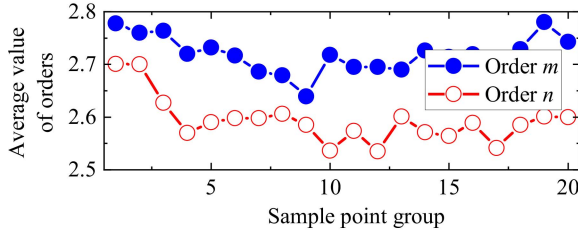
The speed reference is changed from 500 to 2000 r/min⁻¹ at first and decreased to 500 r/min⁻¹ continuously, and the experimental waveforms during the changing reference process are shown in Fig. 8(c), in which it includes the enlarged waveforms of changing reference from 1000 to 1500 r/min⁻¹, and the corresponding average value of orders are shown in Fig. 8(d). As shown in the figure, the proposed method can operate following its reference successfully, and finish this process within about 1.6 s. The average values of orders m and n are both decreased obviously.

Let the time series-based MF-PCC in [24] with fixed orders be named as the conventional MF-PCC, and its experimental waveforms during the same transient-states are shown in Fig. 9. For the previous transient state, the settling time and dropped depth are 6.1 s and 9.42%, respectively, and finishes the latter transient state within about 2.0 s.

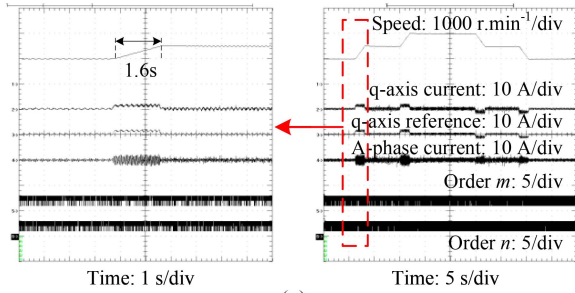
The accumulated error of the q -axis current component is shown in Fig. 10 for these two strategies during continuous 2500 sampling points of the steady state, in which the locations of the current components within 50 sampled points are shown in Fig. 11. The error of AULTS model and its model accuracy are both improved compared to the model in the conventional method. The improvement is 11.94% at the 1000th sampling point, 11.27% at the 2000th sampling point, and 10.41% at the 2500th sampling point. According to the locations of the current components, most of the covering ranges coincide. The range of



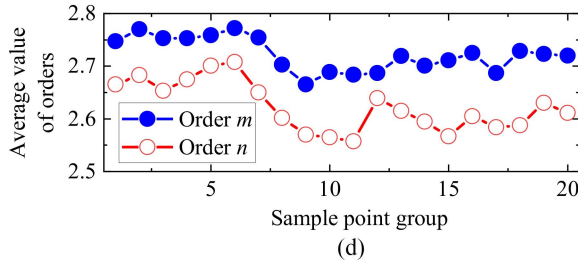
(a)



(b)



(c)

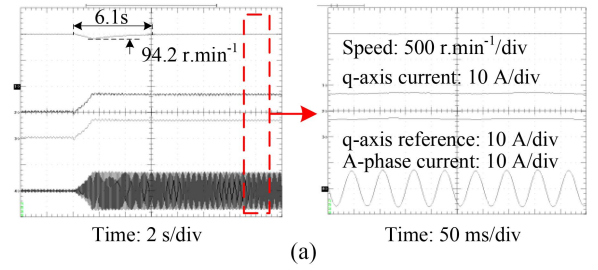


(d)

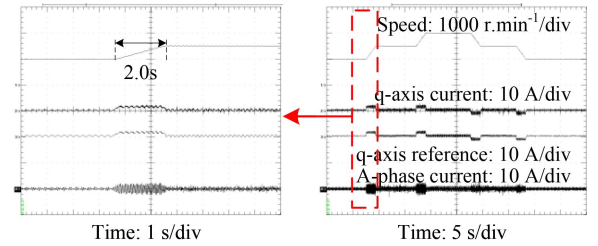
Fig. 8. Experimental waveforms of the proposed MF-PCC strategy. (a) Subject to a load torque disturbance. (b) Corresponding average value of orders during disturbance rejection. (c) Changing speed reference process. (d) Corresponding average value of orders during changing reference.

the proposed MF-PCC is obviously shrunk on the d - and q -axis, i.e. the ripples on these components are both improved compared with the conventional MF-PCC strategy.

For the current quality, the continuous Fourier analysis results of the phase current within 1000 periods for the abovementioned two strategies and the CCS-PCC strategy are shown in Fig. 12(a), and the typical Fourier analysis results are shown in Fig. 12(b). Since some typical harmonic contents are reduced, such as fifth and seventh harmonics, better current quality is obtained by the proposed method compared to the other two strategies. The covering ranges are 0.68%, 0.88%, and 0.91%, respectively,



(a)



(b)

Fig. 9. Experimental waveforms of the conventional MF-PCC strategy. (a) Subject to a load torque disturbance. (b) Changing speed reference process.

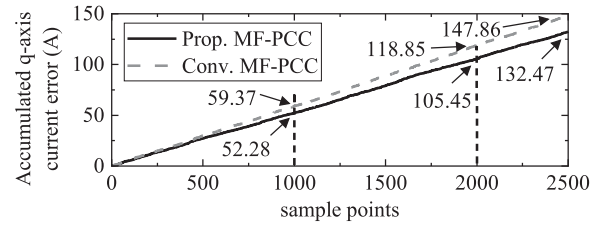


Fig. 10. Accumulated error of the q -axis current component.

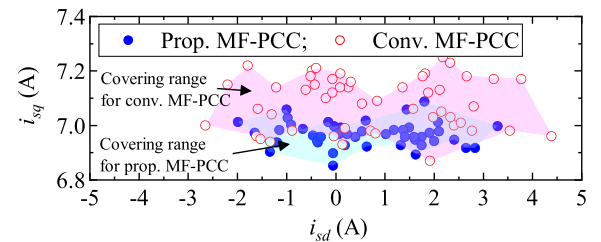


Fig. 11. Locations of current components.

for the compared three strategies, and their median values are about 3.40%, 3.56%, and 3.64%. The realized values by the proposed method are both the minimum in the comparison, and the realized current quality is more stable.

A comparison of some main performance indexes between the proposed and conventional MF-PCC strategies is expressed in Table III, in which the current component covering ranges are roughly computed by the areas in red and blue in Fig. 11. Their dynamics are similar due to the same speed controller. Because of the reduced accumulated error of 10.41%, a better model accuracy is obtained. Based on the better model accuracy, according to median THD of 4.49%, THD range of 22.73%, and

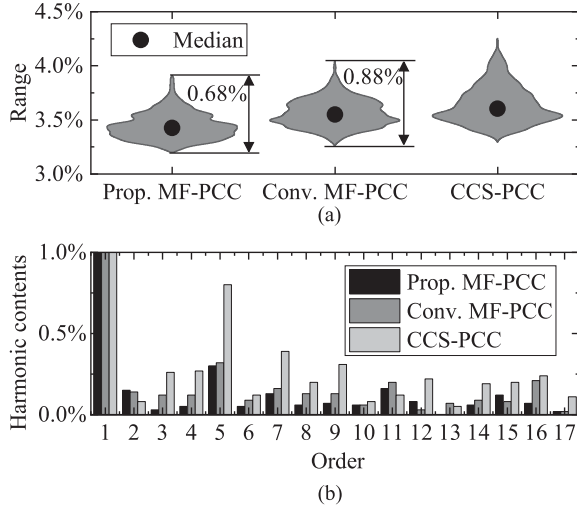


Fig. 12. Phase current quality analysis for the three control strategies. (a) Continuous FFT analysis results. (b) Typical FFT analysis results.

TABLE III
MAIN CONTROL PERFORMANCE INDEXES COMPARISON

Control performance	Prop. MF-PCC	Conv. MF-PCC	Performance increase
Settling time during the disturbance project	5.7 s	6.1 s	—
Dropped depth during the disturbance project	9.28 %	9.42 %	—
Settling time during the changing reference	1.6 s	2.0 s	—
Accumulated q -axis current error	132.47 A	147.86 A	10.41 %
Median THD for phase current	3.40 %	3.56 %	4.49 %
THD range for phase current	0.68 %	0.88 %	22.73 %
Current component covering ranges	0.579 A ²	1.275 A ²	54.56 %

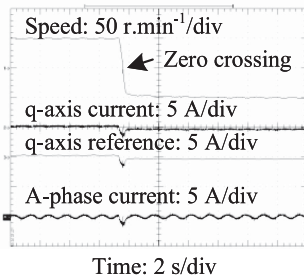


Fig. 13. Experimental waveforms at low-speed operation.

covering range of 54.56%, the current quality and static performance of the proposed MF-PCC are both obviously improved compared with the conventional strategy.

B. Wide Speed Range Validation

To explore the performance at low-speed operation, a step speed reference is set from 50 to -50 r/min, and the experimental waveforms are shown in Fig. 13, under the conditions of no load. Due to the fact that the experimental platform adopts a 2500-line encoder, although the phase current frequency experiences a decrement, the system can operate stably and smoothly turn to reversal rotation.

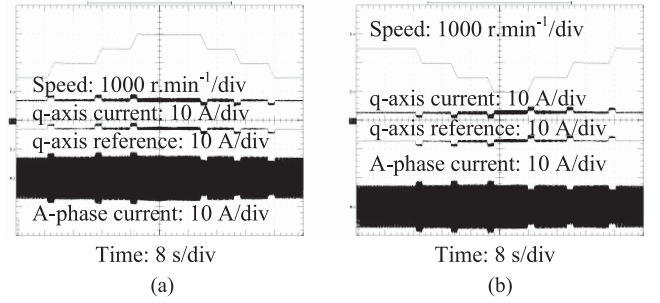


Fig. 14. Experimental platform within a wide speed range. (a) Positive rotation. (b) Reverse rotation.

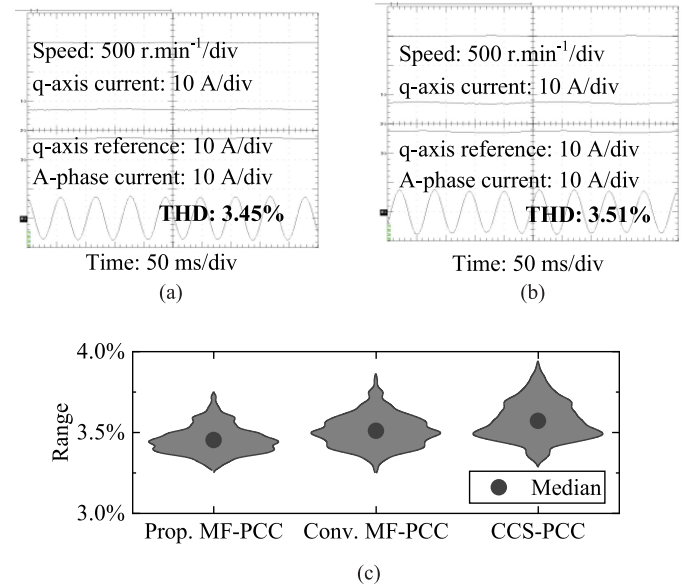


Fig. 15. Experimental waveforms under parameter mismatch condition I. (a) Proposed MF-PCC. (b) Conventional MF-PCC. (c) Continuous FFT result.

Furthermore, the experimental results of continuous acceleration and deceleration tests with positive and negative rotation and a load torque of 11.5 N.m are shown in Fig. 14. They show that regardless of whether it is positive or negative rotation, the system can stably follow the reference and resist various disturbances within a wide speed range.

C. Robustness Validation

Let us define R_{se} , L_{se} , and ψ_{me} as the adopted values of R_s , L_s , and ψ_m in the controller, and different ratios of R_{se}/R_s , L_{se}/L_s , and ψ_{me}/ψ_m are designed to verify the robustness, in which L_{se}/L_s is operated on the d - and q -axis components at the same time. Setting a parameter mismatch condition I as $R_{se}/R_s = 0.1$, $L_{se}/L_s = 0.5$, $\psi_{me}/\psi_m = 2$, the experimental waveforms of the proposed and conventional MF-PCC strategies are shown in Fig. 15(a), and their continuous FFT results are shown in Fig. 15(b) combining with the result of CCS-PCC. It shows that their current qualities are similar to the results in

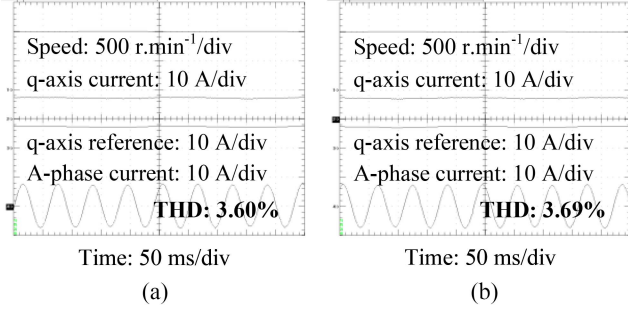


Fig. 16. Experimental waveforms under parameter mismatch condition II. (a) Proposed MF-PCC. (b) Conventional MF-PCC. (c) Continuous FFT result.

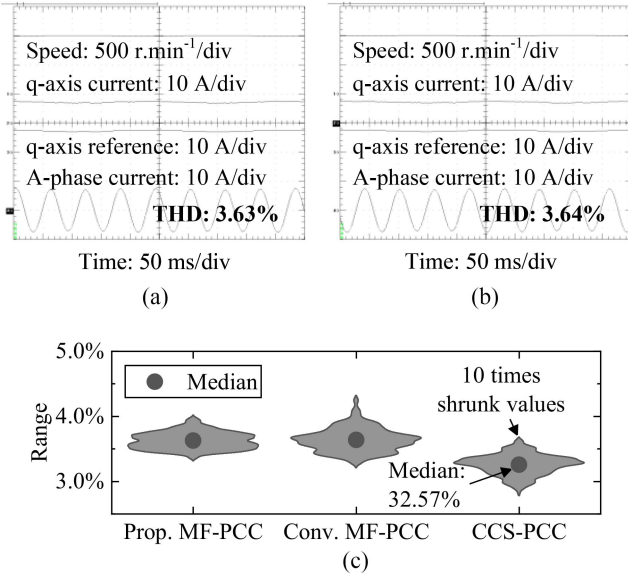


Fig. 17. Experimental waveforms under parameter mismatch condition III. (a) Proposed MF-PCC. (b) Conventional MF-PCC. (c) Continuous FFT result.

Fig. 12(a) since they are not sensitive to this group of mismatch conditions.

The parameter mismatch conditions II and III are defined as $R_{se}/R_s = 1$, $L_{se}/L_s = 2$, $\psi_{me}/\psi_m = 2$, and $R_{se}/R_s = 10$, $L_{se}/L_s = 2$, $\psi_{me}/\psi_m = 0.7$, respectively, and the experimental waveforms are shown in Figs. 16 and 17, in which the results from the CCS-PCC strategy have been shrunk by ten times due to their very large values and difficulty observing. They show that both current qualities of MF-PCCs are not affected by the influences caused by mismatches basically compared with the CCS-PCC with the a priori model, and the best THD is obtained by the proposed method due to its improved model accuracy.

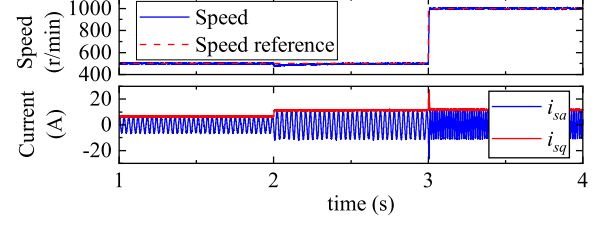


Fig. 18. Simulation results of the proposed method.

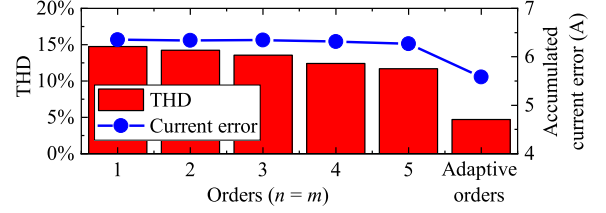


Fig. 19. THD and accumulated current errors by simulation results with different order selections.

D. Auxiliary Simulation Validation

In order to evaluate the advantages of the proposed adaptive order system, a simulation was implemented on MATLAB software, with an increased upper bound of orders set to 5. The load torque and speed reference are changed at 2 s and 3 s, respectively, and the simulation results are shown in Fig. 18. The system can resist the disturbance and track the reference successfully.

The simulation results in Fig. 19 show THDs of the phase current and accumulated current errors in steady state. It is observed that when the model orders are fixed, both THD and error gradually decrease with increasing orders. With the adoption of the adaptive law in the proposed method to select model orders, THD and error exhibit a significant reduction. However, the significant reductions in THD and error cannot be clearly observed in the experimental results in Fig. 12. This may be due to the limited upper bound on the model orders in experiments, which constrains the model accuracy and current quality. These results suggest that the proposed method has the potential to further enhance its performance with an advanced processor with more resources.

V. CONCLUSION

The improved MF-PCC using AULTS that is proposed in this article is capable of effectively upgrading the model accuracy and current quality without the influences caused by parameter mismatches. The coefficients in the model are online estimated by the RLS algorithm, and the model orders are autoselected by the adaptive law to maintain the high fitting for the system. It is worth highlighting that all existed and implied physical parameters are eliminated and the realization is simplified by the ultralocalized model and optimized gain PSO algorithm. Experimental results demonstrate the effectiveness of the proposed AULTS model and its adaptive orders, which have reduced

accumulated error by 10.41% compared with the conventional MF-PCC. In addition, it has also reduced the range and median of current quality deterioration by 4.49% and 22.73%, respectively, highlighting its advantages in providing stable and high-quality current. The appropriate robustness of this method is also verified in the case of parameter mismatch. Furthermore, the proposed method has enough potential to further improve its performance by increasing the upper limit of orders and by using advanced optimization techniques with faster convergence and easier implementation.

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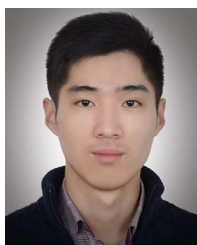
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